

Robotics Overview

Robotics combines mechanical engineering, electronics, control systems, and computer science. Robots use sensors to perceive their surroundings and actuators to produce movement.

Sensors and Perception

LiDAR measures distance using laser light and helps robots build maps. Ultrasonic sensors estimate distance using sound waves. Cameras provide image data for computer vision. An IMU measures angular velocity and linear acceleration, while wheel encoders measure wheel rotation.

Localization and Navigation

SLAM allows a robot to build a map while estimating its own position. Path planning determines a route from a starting point to a goal. Obstacle avoidance helps a mobile robot navigate around objects.

Robot Control

A PID controller uses proportional, integral, and derivative terms to regulate a system. The proportional term responds to current error, the integral term accumulates past error, and the derivative term responds to the rate of change of error.

ROS and Robot Software

Robot Operating System (ROS) provides software libraries and communication tools for robot applications. Nodes perform tasks, topics carry messages between nodes, and services support request-and-response communication.

Power and Safety

Battery capacity and power consumption affect a mobile robot's operating time. Robots should use suitable sensors, collision detection, and safe motion limits around people. Sensor readings may contain noise and should be interpreted carefully.